TRACK 1: ADVANCED CRYPTOGRAPHY SUBMISSION

QuLattice-Decoy

A 4-Layer Post-Quantum Hybrid Cryptosystem with 10-Channel Decoy Routing Logic

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Post-Quantum Vulnerability & Architecture

The Quantum Threat

Shor's Algorithm threatens RSA and ECC infrastructure globally.

Lattice Vulnerabilities

Standard lattice schemes remain vulnerable to Side-Channel Attacks.

Our Solution

Couples LWE primitives with 10-channel dummy-balanced routing.

Mathematical Foundation: LWE Primitive

Additive Key Distribution

Master secret key partitioned across four independent environments with lattice dimension $n = 128$, modulus $q = 2063$, and Gaussian error $\sigma = 3.0$.

$$s_{\text{master}} = \sum_{i=1}^{4} s_i \pmod{q}$$

Encryption & Decryption Mechanism

Ciphertexts are evaluated via error correction holding deterministically with failure probability $P_{\text{fail}} < 2^{-40}$.

$$u = (A^T r + e_1) \pmod{q}$$

System Architecture: 4-Layer Pipeline

HW1 (Root)	Hardware security token / smartcard (s_1).
SW1 (OS)	High-level platform credential share (s_2).
SW2 (Crypto)	Dynamic lattice computation share (s_3).
HW2 (Enclave)	Isolated secure hardware element (s_4).

10-Channel Decoy Multi-Path Routing

DPA Countermeasure

Runtime execution fans out across 10 concurrent paths. 9 decoy channels perform identical matrix multiplications using randomized dummy vectors.

$$C = \{c_0, c_1, \ldots, c_9\}, \quad \exists c_{\text{true}} \in C$$

Constant Hamming Weight

Flattens power trace differentials ($\Delta P \approx 0$).

Strict Volatility

Full vector reconstruction occurs strictly inside HW2 volatile boundary.

Experimental Verification (10,000 Bits)

Metric	Value
Processed Bit Count	10,000
Total Execution Time	9.11 seconds
Bit Error Rate (BER)	0.0000%
Decryption Accuracy	100.0000%

CONCLUSION

Eliminating Single-Point Side-Channel Vulnerabilities

QuLattice-Decoy successfully integrates 4-layer additive secret sharing with 10-channel decoy routing for robust post-quantum encryption.

Thank You | Open for Questions